Hall Ticket Number:

MR-21

Code: B414CS5S2

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. IV Semester Supplementary Examinations, July/Aug-2024

Computer Organization and Architecture

(Computer Science and Engineering)

Time: 3 Hours | ***Answer ALL Questions*** | Max. Marks: 70

Part-A (10 x 2 =20 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
1. a)	Draw the block diagram of digital computer.	2	2	2	1
1. b)	What is the difference between Computer Organization and Computer Architecture?	2	4	2	2
1. c)	List the different addressing modes.	2	1	5	1
1. d)	Discuss about any two types of instruction formats.	2	2	4	2
1. e)	Give the example to represent floating point numbers.	2	2	5	1
1. f)	What is circular shift? Explain with example.	2	3	5	2
1. g)	Define the Read Only Memory. What is its use?	2	1	3	1
1. h)	How does SRAM differ from ordinary DRAM?	2	3	3	2
1. i)	How a vector processor operation is different from conventional scalar processor?	2	4	1	5
1. j)	Differentiate tightly coupled and loosely coupled systems.	2	4	1	5

Part-B (5 x 10 = 50 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
2. a)	Explain different phases of instruction cycle with a neat diagram.	5	2	4	1
2. b)	Draw and discuss about the block diagram of register transfer.	5	2	4	2
	OR				
2. c)	Design a 4-bit combinational decrementer using 4 full adder circuits.	5	3	2	2
2. d)	With neat diagram, explain the operation of input-output configuration in digital computer.	5	2	2	3
3. a)	Demonstrate the addressing modes in digital computer with examples.	5	3	3	2
3. b)	Explain the various instruction formats with an example.	5	2	3	1
	OR				
3. c)	Explain about address sequencing capabilities in control memory.	5	2	3	4
3. d)	Discuss the elements of micro-programmed control organization.	5	2	3	1
4. a)	Compare signed and unsigned binary numbers with table.	5	4	5	2
4. b)	Perform the arithmetic operation (+41)+(-13) and (-41)-(-13) in binary using signed 2's complement representation for negative numbers.	5	3	5	3
	OR				
4. c)	Explain decimal arithmetic operations with examples.	5	2	5	1
4. d)	Solve a step-by-step multiplication of (+15) X (-13) using Booth's algorithm.	5	3	5	11
5. a)	Discuss the Memory hierarchy with a neat diagram.	5	2	3	1
5. b)	Explain about Secondary or Auxiliary Memory.	5	2	3	1
	OR				
5. c)	Draw the block diagram of a typical DMA controller and explain.	5	3	3	1
5. d)	Explain about isolated and memory mapped I/O.	5	2	3	1
6. a)	Explain about arithmetic pipeline with an example.	5	3	4	5
6. b)	What is the Inter processor Synchronization? Explain the Synchronization techniques.	5	2	1	5
	OR				
6. c)	Explain various interconnection structures of a multiprocessor system with neat diagrams.	5	2	1	6
6. d)	Discuss the working and the importance of RISC architecture.	5	2	1	1

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome